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ORIGINAL RESEARCH PAPER

Micronized palmitoylethanolamide reduces joint pain and glial cell activation

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Abstract

Objective and design Temporomandibular disorder (TMD) is a common painful condition in the temporomandibular joint (TMJ). Joint inflammation is believed to be a chief cause of pain in patients with TMD, through the release of pro-inflammatory cytokines that induce peripheral sensitization of nerve terminals followed by microglial stimulation.

Materials and subject TMJ was induced in rats with the injection of complete Freund's adjuvant (CFA) emulsion into the left TMJ capsule.

Treatment The present study would assess the effects of micronized palmitoylethanolamide (m-PEA) on glial activation and trigeminal hypersensitivity.

Methods Ten mg/kg m-PEA or corresponding vehicle was administered 1 h after CFA and mechanical allodynia and edema were evaluated at 24 and 72 h after CFA injection.

Results CFA-injected animals showed TMJ edema and ipsilateral mechanical allodynia accompanied by a robust growth in GFAP protein-positive satellite glial cells and activation of resident macrophages in the TG. Moreover, m-PEA administration significantly reduced the degree of TMJ damage and pain, macrophage activation in TG and up-regulation of Iba1.

Conclusions The results confirm that m-PEA could represent a novel approach for monitoring pain during trigeminal nerve sensitization.

Keywords Palmitoylethanolamide · Temporomandibular joint · Inflammation · Glia · Micronization

Introduction

Osteoarthritis (OA) is a complex and highly prevalent disease affecting about 10% of US adult population [1] and a primary reason of disability [2]. An increasing evidence indicates a crucial role of inflammatory stimuli in the pathogenesis and progression of OA. Among the clinical problems associated to OA [1], pain is markedly important: it involves both neuropathic and nociceptive components and it is linked to local and central sensitization and therefore, also to the possible development of chronic pain [3].

OA commonly affects the temporomandibular joint (TMJ), often leading to severe pain and to a limitation of the jaw range of motion. Persistent non-resolving-TMJ inflammation is consequent to a dysregulation of neuron–non-neuronal cell interaction [4–6]. The satellite glial cells surrounding the trigeminal ganglion (TG) neurons as well as microglia and astrocytes in trigeminal spinal

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nucleus caudalis actively participate in the information processing of sensory signals from afferent terminals to the spinal cord and in the cortical experience of pain. Changes in this communication following noxious stimuli is known to lead to abnormal pain conditions [7, 8]. The involvement of ganglia satellite glial cells and microglia in medullary and cervical dorsal horn in TMJ pain have been recently confirmed also in the adjuvant-induced arthritis model [9].

The use of compounds that relieve inflammation process might limit the development of peripheral and central sensitization, avoiding the progression from acute to chronic pain and the worsening of OA. A new pharmacological strategy to restore physiological glia functions includes fatty acid ethanolamines (FAEs), a promising family of naturally occurring lipid mediators formed on demand to decrease inflammation and restore tissue homeostasis. The principal members of this family are the endocannabinoid *N*-arachidonoylethanolamine (anandamide) and its congeners *N*-stearoylethanolamine, *N*-oleoylethanolamine and *N*-palmitoylethanolamine (PEA). Among these, PEA has emerged as an innovative therapeutic strategy to control the mast cell–microglia axis, showing anti-inflammatory, pain-relieving and neuroprotective properties. In that regard, PEA showed great efficacy in the treatment of pain and inflammation. Several of previous works demonstrated the beneficial effects of PEA alone and in combination in different models of inflammation and pain [10, 11], and in a mouse model of CIA [12]. The exact mechanism of action of PEA is not well known although PEA could interact with peroxisome proliferator-activated receptor (PPAR)-α, which are involved in its anti-inflammatory effects and with the orphan G-protein-coupled receptor, GPR55 [13]. Interestingly, previous studies also indicated that PPAR-δ and PPAR-γ can contribute to the anti-inflammatory activity of PEA [14].

Moreover, a recent study investigated the effects of a micronized formulation of PEA (m-PEA, Normast 300) *vs* ibuprofen on temporomandibular joint (TMJ) pain, whereby a greater enhancement in pain relief and range of motion in the PEA group was associated to long-lasting efficacy. In addition, the observed correlation between inflammatory pain and activation of glial cells in the TG and central nervous system [9] suggests that glial cells could be the pharmacological targets of molecules with pain-relieving effects, such as m-PEA. m-PEA is obtained through a micronization technique to achieve microparticles (< 10 µm), that increased surface area and rate of dissolution, together with a reduced variability of absorption. Based on these evidence, here we assessed the properties of m-PEA on glial cell activation following induction of TMJ inflammation and trigeminal hypersensitivity.

Materials and methods

Animals

The study was conducted using Sprague–Dawley male rats (200–230 g, Envigo, Italy). The animals were fed standard laboratory chow and allowed free access to water in an air-conditioned room with a 12-h light/12-h dark cycle.

All procedures performed in animals were in accordance with the ethical standards of the Italian regulations on the protection of animals used for experimental and other scientific purposes (D.M.116192) as well as with EEC regulations (O.J. of E.C. L 358/1 12/18/1986).

Induction of TMJ inflammation

To achieve the activation of ganglia satellite glial cells and spinal microglia, the induction of TMJ inflammation was performed as previously reported [9, 15–17].

Experimental groups

Rats were randomly divided into the following groups:

Sham group Rats received an equal volume of saline into the left TMJ capsule using the same technique.

Sham + mPEA group Rats received m-PEA administered intraperitoneally (ip) at a dose of 10 mg/kg;

CIA group Rats were subjected to TMJ inflammation by administering 50 µl of CFA oil/saline (1:1) emulsion into the left TMJ capsule;

CIA + mPEA group Rats were subjected to TMJ as described above and were treated by ip administration of m-PEA at a dose of 10 mg/kg 1 h post-CFA.

Rats were killed, respectively, at 24 and 72 h after CFA or saline injection. Only male rats were used because they present a higher nociceptive response to CFA-induced inflammation [18], probably due to an anti-inflammatory effects of estrogens in female rats.

The minimum number of mice for every technique was estimated with the statistical test a priori power studies with the G-power software, this statistical test supplies a professional method to analyze the sample size required to make the experiments.

Micronized PEA preparation and treatment

m-PEA was provided by Epitech Group SpA. m-PEA was resuspended in 5% Pluronic F-68 to preserve particle size distribution of the lipidic molecule. m-PEA was administered ip at a dose of 10 mg/kg 1 h post-CFA or

saline. Sham and CFA animals received the vehicle used for m-PEA.

Behavioral test

Mechanical allodynia was measured using a previously described protocol with minor modifications [16].

Measurement of Evans blue dye extravasation

Evans blue dye was injected into the tail vein [19] 10 min before perfusion (see below). The left TMJ was dissected out, cut into small blocks and incubated overnight at room temperature in a 7:3 (v/v) mixture of acetone and 35.2 mM sodium sulfate on a shaking table. Samples were then centrifuged, the supernatant collected and dye absorbance determined spectrophotometrically at 620 nm.

Tissue processing for histological experiments

Twenty-four and 72 h after CFA injection, rats were killed and both brain and TG were excised and fixed in 10% (w/v) formaldehyde in phosphate-buffered saline (PBS). Eight-micrometer sections were prepared from paraffin-embedded tissues.

Immunofluorescence for GFAP/NeuN, Iba1 and ED1

After deparaffinization and rehydration the detection of glial fibrillary acidic protein and neuron-specific nuclear protein (GFAP/*NeuN*), of the number and of the activation of macrophages phenotype (Iba1 and ED1) were carried out in ophthalmic (V1), maxillary (V2) and mandibular (V3) TG divisions after boiling in 0.1 M citrate buffer for one minute. Sections were incubated with polyclonal rabbit anti-GFAP (1:100, vol/vol Santa Cruz Biotechnology), polyclonal mouse anti-Iba1 (1:100, vol/vol Santa Cruz Biotechnology), polyclonal mouse anti-ED1 (1:100, vol/vol Santa Cruz Biotechnology) and mouse anti-NeuN (1:100; vol/vol Santa Cruz Biotechnology) antibodies in a humidified oxygen and nitrogen chamber over night at 37 °C. Sections were incubated with a secondary antibody fluorescein isothiocyanate (FITC)-conjugated anti-mouse Alexa Fluor-488 antibody (1:2,000 vol/vol Molecular Probes, Monza, Italy) and with TEXAS RED-conjugated anti-rabbit Alexa Fluor-594 antibody (1:1000 in PBS, vol/vol Molecular Probes, Monza, Italy) for 1 h at 37 °C. For nuclear staining, 2 µg/ml 4′,6′-diamidino-2-phenylindole (DAPI; Hoechst, Frankfurt, Germany) in PBS was added. Sections were observed at 20× magnification using a Leica DM2000 microscope (Leica, Milan, Italy).

Quantitative analysis of immunopositive cells in the TG was carried out by using the Image-J software. The number of positive cells in TG was calculated on sections acquired at 10× magnification. The perineuronal regions of the V1, V2 and V3 division were traced, and the number was calculated. The number of positive cells has been normalized to the area of measurement.

Immunohistochemical localization of GFAP and Iba1

After deparaffinization, endogenous peroxidase was quenched with 0.3% (v/v) hydrogen peroxide in 60% (v/v) methanol for 30 min. Brainstem and spinal cord sections (three sections for each sample) were permeabilized with 0.1% (w/v) Triton X-100 in PBS for 20 min. Non-specific adsorption was minimized by incubating the section in 2% (v/v) normal goat serum in PBS for 20 min. Endogenous biotin or avidin binding sites were blocked by sequential incubation for 15 min with biotin and avidin (Santa Cruz, Milan, Italy), respectively. Sections were incubated overnight with rabbit anti-GFAP (1:100; Santa Cruz Biotechnology) or rabbit anti-Iba1 (1:100; Santa Cruz Biotechnology). Sections were then washed with PBS and incubated with secondary antibody. Specific labeling was detected with a biotin-conjugated goat anti-rabbit IgG and avidin–biotin peroxidase complex. The counter stain was developed with diaminobenzidine (brown) and nuclear fast red (red background). To confirm the specificity of GFAP immunoreactivity, some sections were incubated with anti-GFAP antibody in the presence of excess GFAP (10 mM). Binding specificity for Iba1 was assessed by incubating with primary or secondary antibody only. No positive immunostaining was seen under these conditions. Acquired immunohistochemical images ($n = 5$) were assessed by densitometry using an imaging densitometer (AxioVision, Zeiss, Milan, Italy) and computer program.

Statistical analysis

All values are expressed as mean ± standard error of the mean (SEM) of n observations. For in vivo studies "*n*" represents the number of animals used.

The results were analyzed by one-way ANOVA followed by a Bonferroni post hoc test for multiple comparisons. A p value less than 0.05 was considered significant.

Results

Evans blue dye extravasation and mechanical allodynia

To evaluate CFA-induced TMJ extravasation, Evans blue dye concentration was measured in TMJ tissues obtained from sham and CFA-injected rats treated with m-PEA or

vehicle. The Evans blue plasma extravasation (PE) in the TMJs of rats injected with CFA was significantly greater as compared to sham animals and sham animals treated with m-PEA, both at 24 h and, to a lesser extent, at 72 h (Fig. 1a). However, PE was significantly reduced in the CFA-injected rats treated with m-PEA both at 24 and 72 h (Fig. 1a). m-PEA did not modify PE in sham animals (Fig. 1a).

Mechanical threshold was determined in sham group before saline or CFA injection into the left TMJ (Time 0; Fig. 1b). Head withdrawal thresholds were measured from the left orofacial regions. CFA-injected rats displayed a significantly minor threshold to mechanical stimuli when compared to baseline value and sham animals; the mechanical threshold reduction was statistically significant starting 24 h post-injection (Fig. 1b). No significant change in mechanical threshold was observed in the sham group, even when treated with m-PEA (Fig. 1b); instead CFA-injected rats treated with m-PEA exhibited less orofacial mechanical allodynia as compared to the CFA-vehicle group (Fig. 1b).

GFAP immunoreactivity following TMJ-induced inflammation

The number GFAP-positive encircled neurons, evaluated by immunofluorescence analysis, was very low in TG of sham animals (Fig. 2c–e). In CFA animals, GFAP signals were

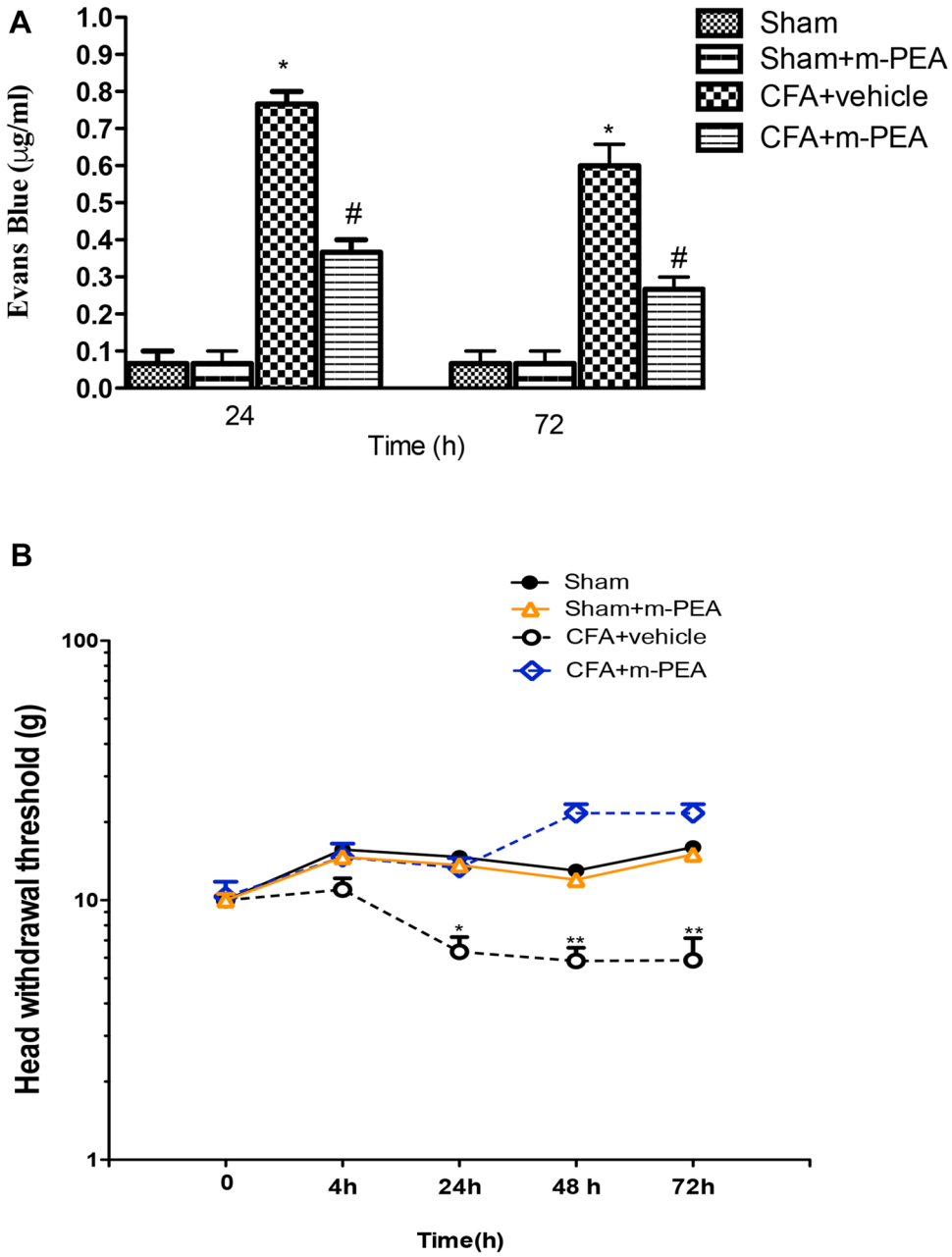


Fig. 1 Effects of m-PEA on CFA-induced-TMJ extravasation and mechanical allodynia. Micronized PEA (m-PEA) reduces CFA-induced TMJ extravasation (**a**) and mechanical allodynia (**b**). 10 mg/kg m-PEA was administered intraperitoneally 1 h post-CFA or saline. Values are expressed as mean ± SEM; $n = 10$ for each group. Rats used for determination plasma extravasation at 72 h were also used to measure the mechanical hyperalgesia. $^{*}p < 0.05$, $^{**}p < 0.01$ vs sham; $^{\#}p < 0.05$ compared to CFA/vehicle rats

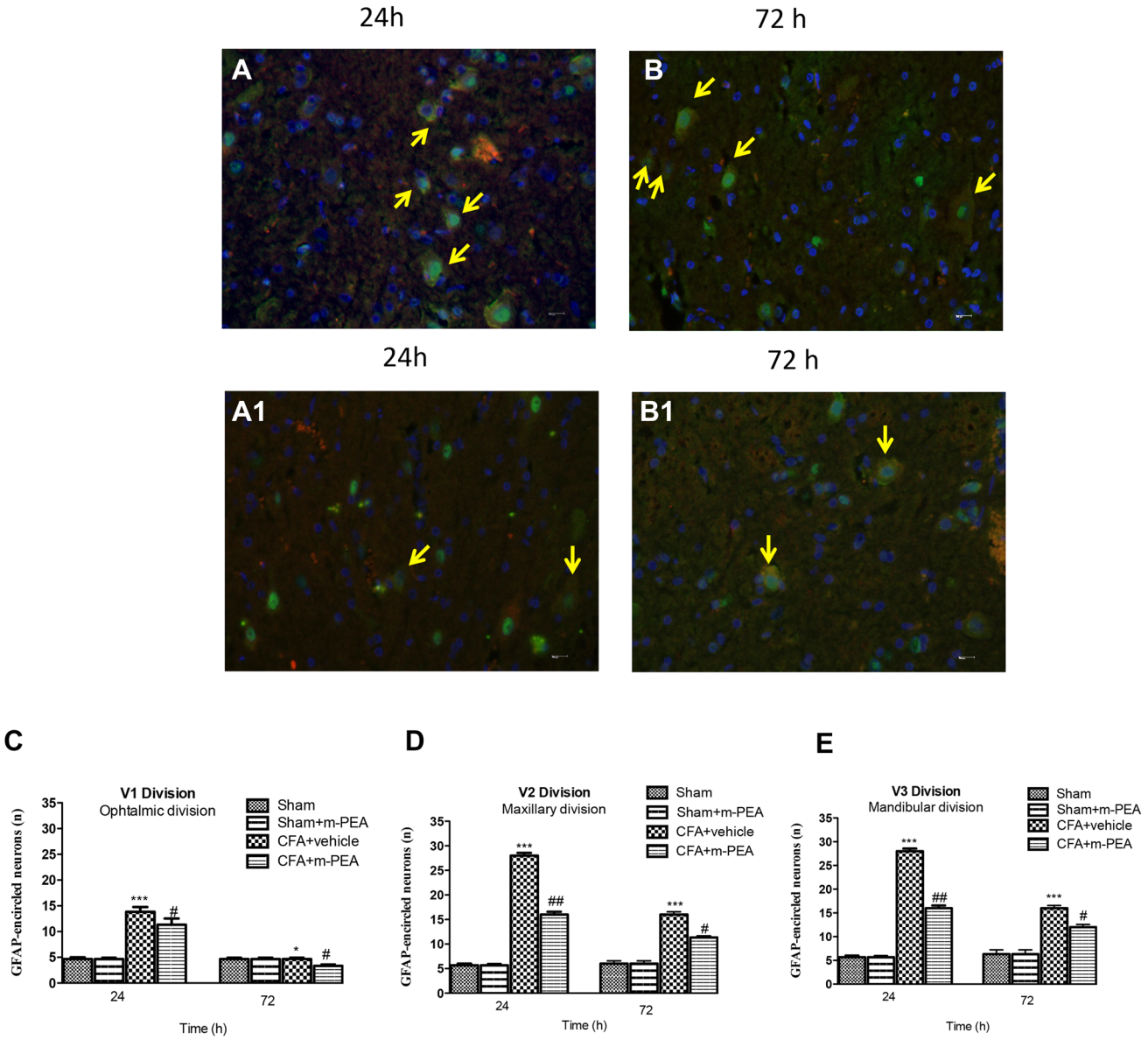


Fig. 2 GFAP-encircled neurons after induction of TMJ inflammation. A significant increase in the number of GFAP was observed in the ipsilateral TG early at 24 and 72 h post-CFA injection (**a, b**) compared to sham (data not shown). Few GFAP + SGCs (red) encircling NeuN + neurons (green) were identified in the m-PEA group compared to vehicle group at 24 and 72 h post-CFA (**a1, b1**). The yellow arrows (in the representative images) indicate the GFAP-encircled neurons relative to Division V2, maxillary division. Scale bars 20 µm. **c**–**e** The number of GFAP was measured in all 3 divisions (V1-ophthalmic, V2-maxillary, V3-mandibular). Values are expressed as mean ± SEM; $n = 10$ for each group. $^{***}p < 0.001$ compared to sham rats; $^{\#}p < 0.05$ compared to the CFA + vehicle; $^{\#\#}p < 0.01$ compared to the CFA + vehicle. (Color figure online)

higher at 24 h in the ipsilateral V3-mandibular, V2-maxillary (Fig. 2a, d, e) and V1-ophthalmic divisions of the trigeminal nerve (Fig. 2c). Such an increase was less pronounced when evaluated at 72 h (Fig. 2b–e). In CFA/m-PEA-treated animals a reduction of GFAP immunoreactivity was observed as compared to CFA/vehicle group, both at 24 h (Fig. 2a1, c, d, e) and 72 h (Fig. 2b1, c, d, e).

In addition, brainstem sections medulla oblongata (Fig. 3a, b1, b2) and spinal cord (Fig. 3c, d1, d2) processed for immunohistochemistry revealed no change of GFAP immunoreactivity both in sham and CFA-injected rats at any time (Fig. 3).

Iba1 as a marker of resident macrophages in TG after TMJ inflammation and brainstem microglia

Relative Iba1 fluorescence intensity showed no significant changes in number of Iba1^{+}-resident macrophages

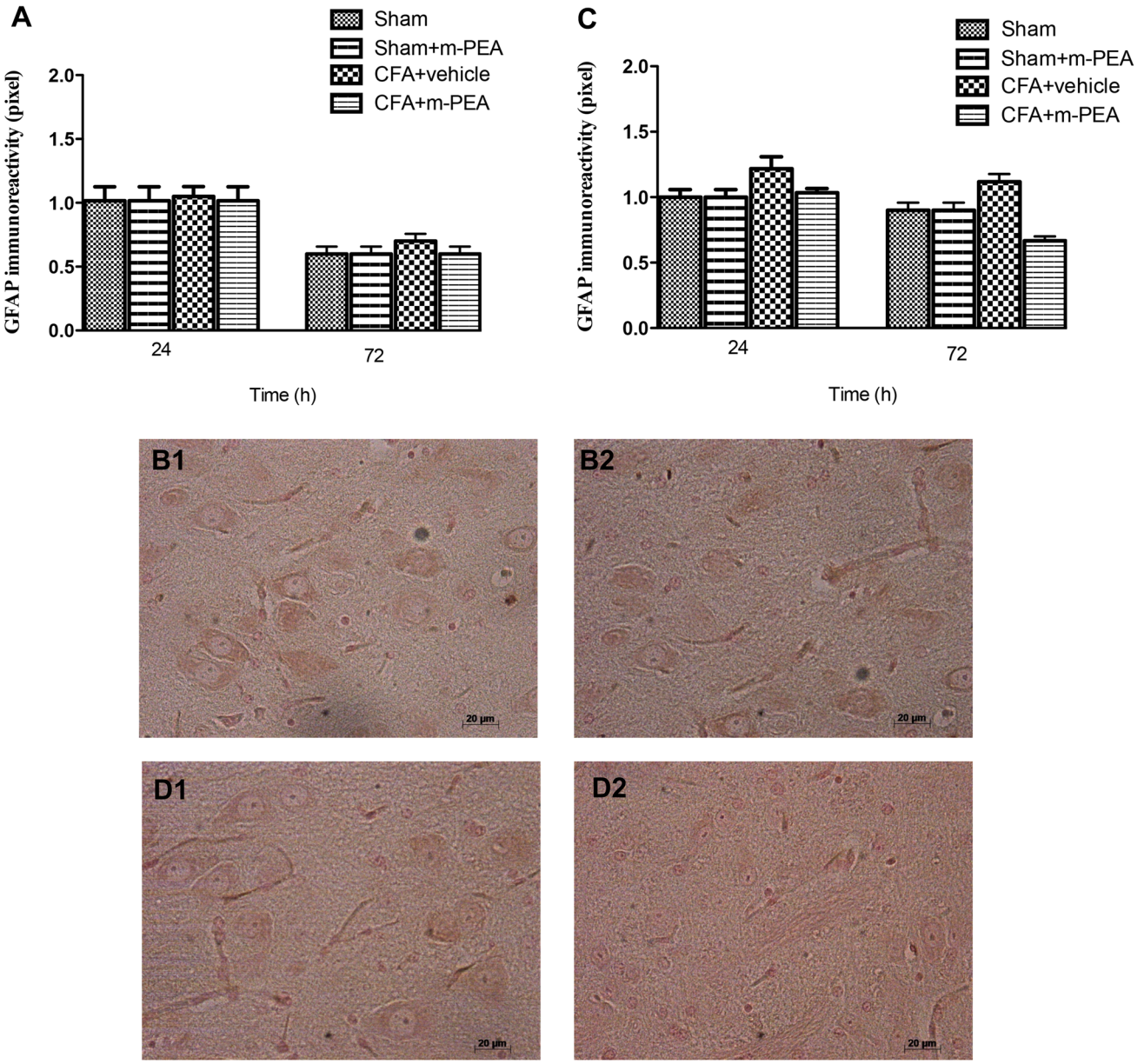


Fig. 3 Astroglial activation in the spinal trigeminal nucleus of medulla oblongata and spinal cord. Densitometric analysis of GFAP immunoreactivity showed no significant difference between CFA-injected rats and m-PEA injected rats at 24 h (**a** and see **b1, b2**) and 72 h (**c** and see **d1, d2**). Values are expressed as mean ± SEM; $n = 10$ for each group

in all divisions of the ipsilateral TG at both 24 and 72 h (Fig. 4a–e). No important variation in the number of Iba1$^+$ cells was also observed in CFA-injected rats, either in the vehicle group or in m-PEA-treated rats (Fig. 4a1, b1, c, d, e). In addition, microglia were observed in the medulla oblongata (Fig. 5a, b1, b2) and spinal cord (Fig. 5c, d1, d2) both at 24 and 72 h after CFA-induced TMJ inflammation.

An up-regulation of the microglia was observed in the medulla oblongata (Fig. 5a, b1) and spinal cord (Fig. 5c, d1) at 72 h principally in the CFA-induced TMJ group compared to sham group and sham treated with m-PEA (Fig. 5a, c). Instead CFA-injected rats treated with m-PEA displayed a reduction of microglia activation both in the medulla oblongata (Fig. 5a, b2) and spinal cord (Fig. 5c, d2) at 72 h as compared to the CFA-vehicle group. No significant change in microglia activation was observed at 24 h either in the trigeminal subnucleus caudalis of medulla oblongata and in the spinal cord (Fig. 5a, c).

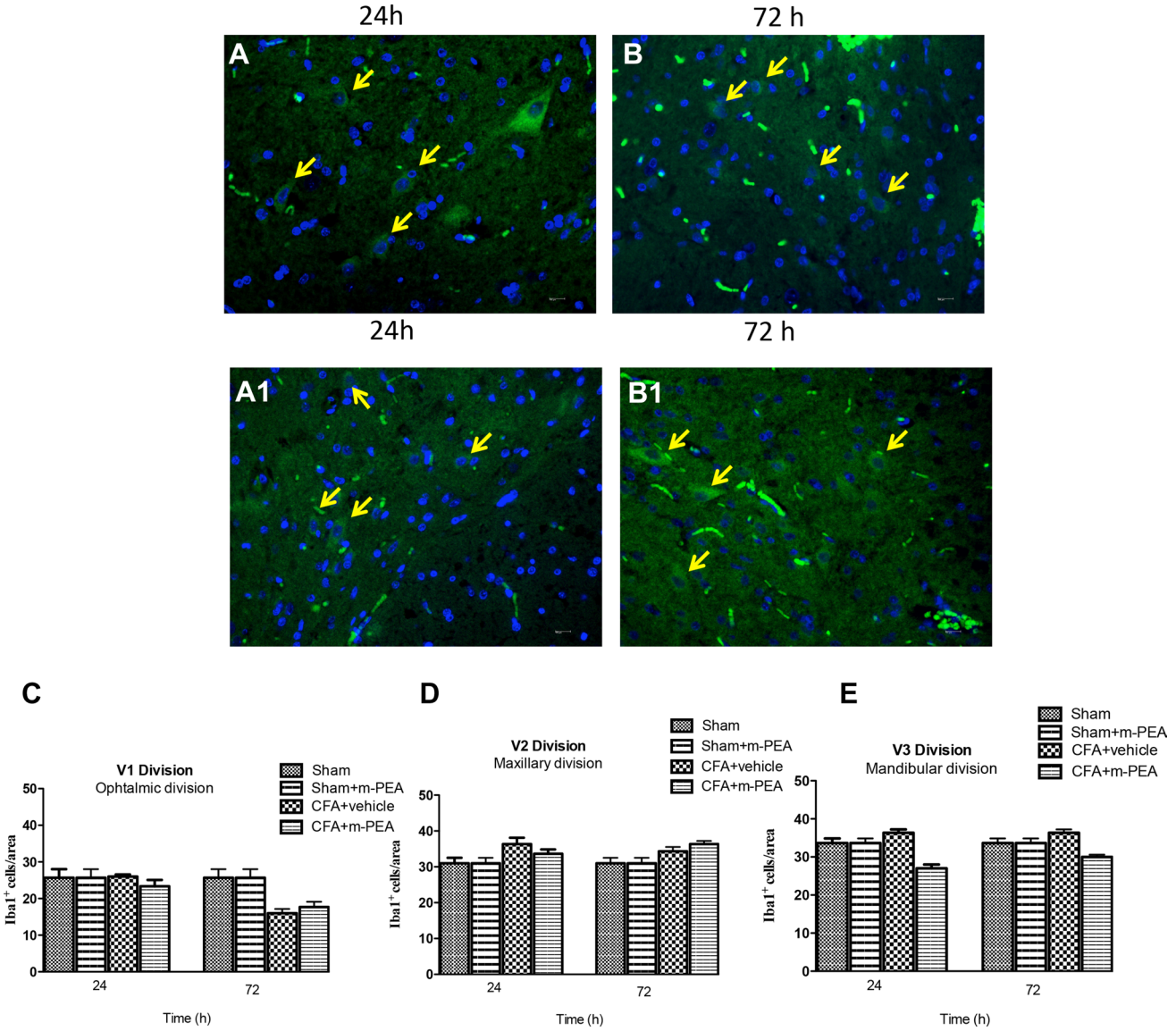


Fig. 4 Iba1-positive cells after induction of TMJ inflammation. The number of Iba1-positive cells (green) showed no significant difference at 24 h (**a, a1, c–e**) or 72 h (**b, b1, c–e**) after CFA injection. Representative images are relative to Division V2, maxillary division. Scale bars 20 µm. The number of Iba1-positive cells (**c–e**) was detected in any of the three TG divisions in all experimental groups. Values are expressed as mean ± SEM; $n = 10$ for each group

Increased ED1 expression as marker of macrophage activation in TG

ED1 immunofluorescence analysis showed clear signs of macrophage activation in the TG.

The number of ED1$^+$ cells significantly increased in the three TG divisions as compared to sham animals and sham treated with m-PEA after CFA injection (Fig. 6a, c–e) especially at 24 h. These effects persisted at 72 h, although to a minor extent (Fig. 6b–e). Treatment with m-PEA significantly reduced ED1-positive cell numbers both at 24 and 72 h (Fig. 6a1, b1, c–e) in all of the three TG divisions.

Discussion

The present study confirms the immunological activation associated with CFA-induced TMJ inflammation: glial cells become activated in both the TG and brainstem following CFA injection [9]. At the time analyzed, as also reported previously [9], there was no alteration in GFAP immunohistochemistry in the spinal trigeminal nucleus in CFA/TMJ animals. In a previous study, a strong GFAP up-regulation has been reported 5 days following CFA injection when evaluated by using western blotting analysis and a less but significant increase was present at 1 and 3 days post-CFA

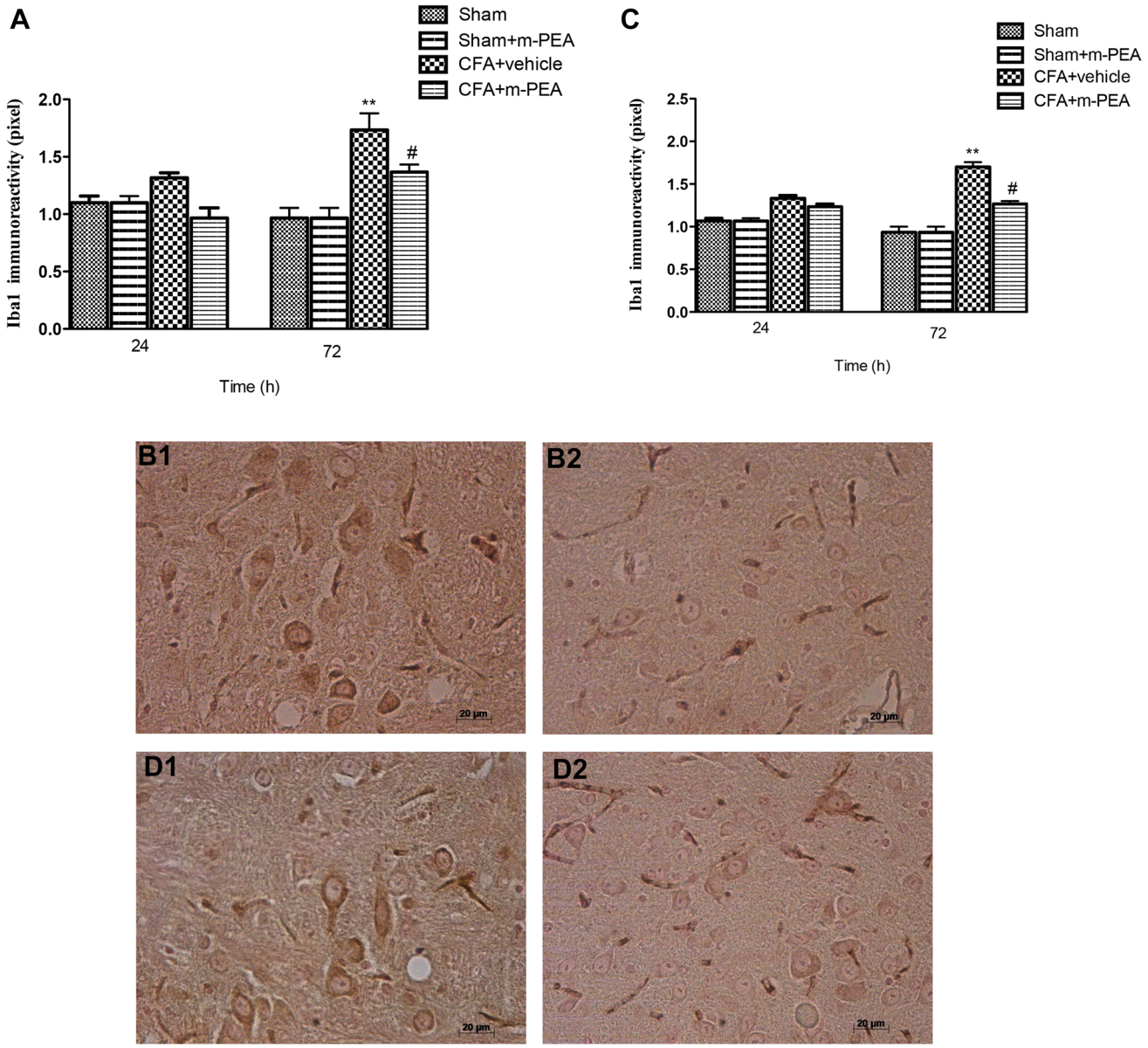


Fig. 5 Microglial activation in the nucleus of medulla oblongata and spinal cord. Densitometric analysis of Iba1 immunoreactivity showed a significant increase in Iba1 reactivity in the CFA + vehicle group of both the trigeminal subnucleus caudalis in the medulla oblongata (**a** and see **b1**) and the dorsal horn of the cervical spinal cord (**c** and see **d1**) 72 h after CFA injection compared to sham and sham treated with m-PEA (data not shown). m-PEA treatment reduced microglial activation both in the t medulla oblongata and in spinal cord (**a, c** and see **b2, d2**). No significant difference was found at 24 h in all experimental groups. Values are expressed as mean ± SEM; $n = 10$ for each group. $^{**}p < 0.01$ *vs* sham; $^{\#}p < 0.05$ vs CFA + vehicle

injection [20]. On the other hand, the temporal expression profile of specific pro-inflammatory cytokines and glial activation in CFA model revealed a role of microglia in the primary phase and astrocytes in maintaining hypersensitivity [21]. For this reason, our data are not in contrast with other orofacial pain models in which an increase of GFAP immunoreactivity has been reported [22, 23].

These data obtained confirm that glial cells may consider as a targets for modulate pain during trigeminal nerve sensitization. Of note, the present results demonstrate that CFA-induced TMJ inflammation and pain were reduced by m-PEA administration, as evidenced by the reduction in edema and mechanical allodynia after TMJ induction. These effects were associated with a reduced density of GFAP-positive cells in the ophthalmic, maxillary and mandibular divisions of the TG, and reduced macrophage activation in terms of ED1 fluorescence intensity and number of $ED1^+$ cells. In the brain, m-PEA counteracts the

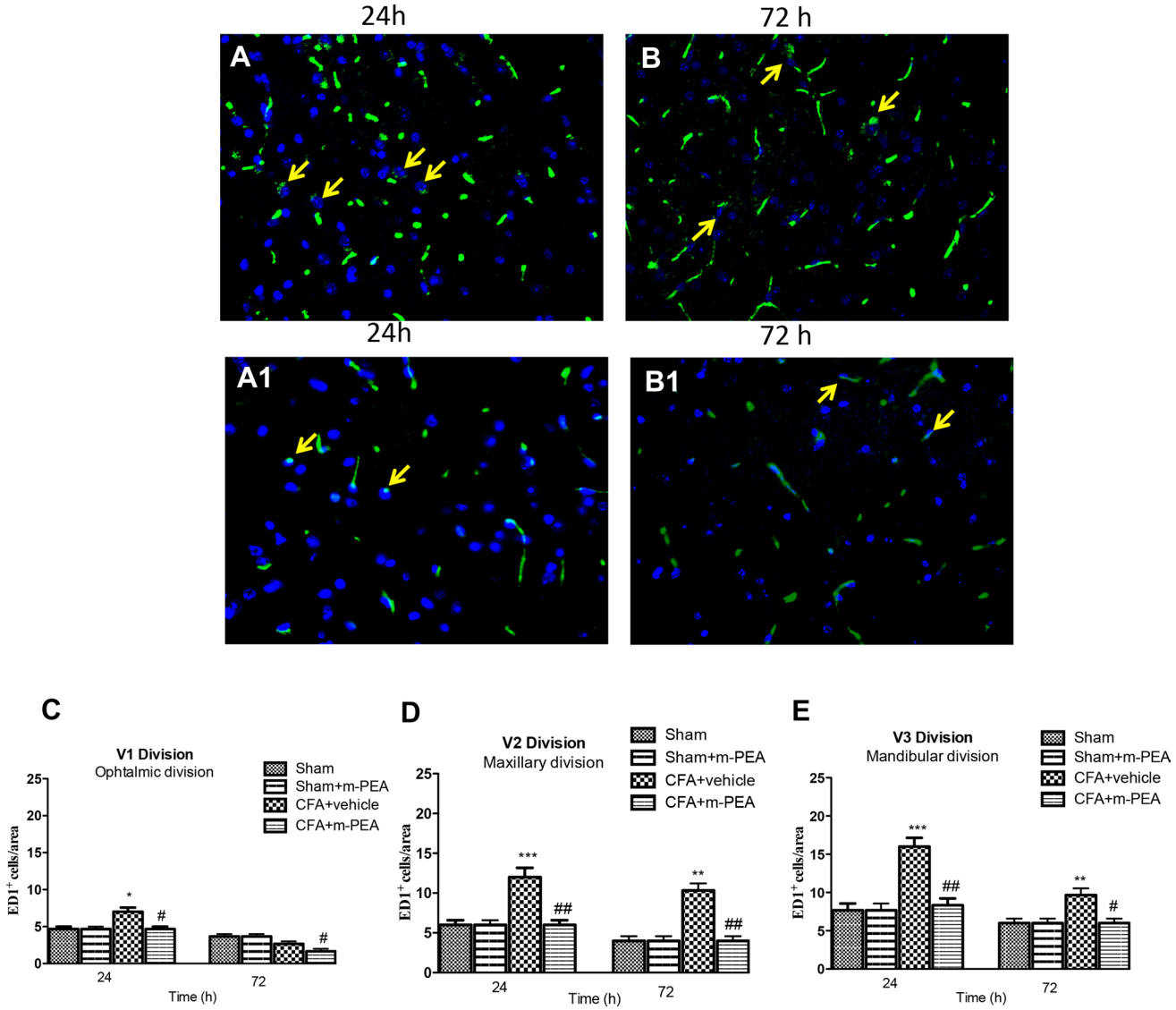


Fig. 6 ED1-positive cells after induction of TMJ inflammation. ED1-positive cells (green) of activated macrophages, showing up-regulation at both 24 and 72 h (**a, b**) post-CFA injection compared to sham (data not shown). m-PEA treatment reduced ED1-positive cells number at both 24 and 72 h (**a1, b1**). Representative images are relative to Division V2, maxillary division. The number of ED1-positive cells was detected in any of the three TG divisions, 24 and 72 h post-CFA (**c**–**e**). Values are expressed as mean ± SEM; $n = 10$ for each group. $^{*}p < 0.05$ vs sham, $^{***}p < 0.001$ *vs* sham; $^{\#}p < 0.05$ vs CFA + vehicle; $^{\#\#}p < 0.01$ vs CFA + vehicle. (Color figure online)

increase in Iba1 immunoreactivity both in medulla oblongata and spinal cord.

Systemic administration of the β-lactam-based *N*-acylethanolamine acid amidase inhibitor ARN726 is reported to reduce CFA-induced edema, confirming the capacity of endogenous PEA to mitigate inflammation [24]. Furthermore, exogenous PEA appears to reduce edema in different experimental models of inflammation and pain [6]. The ability of PEA to normalize mechanical sensitization of peripheral trigeminal nerve endings has been observed also when diethylenetriamine was inoculated into the trigeminal ganglion in vivo [25]: PEA dose-dependently inhibited nitric oxide-induced mechanical sensitization of masticatory muscle nociceptors and prostaglandin E2 release by satellite glial cells. Since PEA administration resulted ineffective on mechanical sensitization induced by intraganglionic injection of prostaglandin E2, it was suggested that the effect of PEA was likely mediated by inhibiting prostaglandin synthesis in satellite glial cells. The involvement of glia in PEA effects is supported also by evidence showing a comparable efficacy between PEA and minocycline in the suppression of PGE2 produced in glial cells [26]. Minocycline treatment significantly increased

PEA levels in the spinal cord of selective spinal nerve ligation rats, compared to contralateral spinal cord [27].

The present results identify the involvement of different non-neuronal cells in m-PEA-induced reduction of mechanical allodynia that include peripheral macrophages and central microglia. These data confirm that m-PEA may also act on central targets to reduce central sensitization consequent to peripheral inflammatory stimulation. On the other hand, the smaller particle size of m-PEA is associated with a higher surface-to-volume ratio which may contribute to better solubility of lipophilic molecules like PEA [11, 28, 29].

The capacity of PEA to reduce both inflammation and neuroinflammation, mainly by regulating peripheral and central non-neuronal cells that result in pain relief, is documented in a growing number of studies. PEA is produced/hydrolyzed by glial cells and mast cells [30–32]. It down-modulates mast cell activation [33, 34], and controls glial cell behaviors [35]. The administration of PEA in micronized and ultramicronized formulations is reported to reduce pain behaviors and to inhibit somatosensory activation in different models of acute and chronic pain [6].

The results obtained here demonstrated an attenuation of trigeminal peripheral ending sensitization along with down-regulation of glial cells and resident macrophage activation by m-PEA treatment, suggesting a possible option for management of TMJ osteoarthritis by modulating the active inflammatory process, preserving function and further deformity and by relieving pain. However, a long-lasting study aimed both at verifying the contribution of the different glial and immunocompetent cells in the different phases of inflammatory response and the ability of m-PEA to control the activation of non-neuronal cells also during chronic TMJ neuroinflammation.

Furthermore, the results obtained also support a potential use of m-PEA for management of craniofacial pain and headache. This idea is supported by the marked improvement in a female patient affected by occipital neuralgia after oral intake of ultramicronized PEA [36], and by a previous study demonstrating that m-PEA was more effective than ibuprofen in the treatment of TMJ inflammatory pain [37]. In this last study, pain intensity progressively decreased in the m-PEA treated group, confirming a lack of tolerance with chronic treatment. The effect of m-PEA in TMJ pain was superior to that achieved with ibuprofen, suggesting that m-PEA could act on the neuropathic component of TMJ pain, which is mainly mediated by persistent activation of non-neuronal cells, in particular brain microglia.

In conclusion, the findings reported here confirm the important contribution of non-neuronal cells in TMJ inflammation and pain and the possible application of m-PEA as an innovative therapeutic agent to decrease inflammation thereby reducing trigeminal hypersensitivity.

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Compliance with ethical standards

Conflict of interest Salvatore Cuzzocrea is co-inventor on a patent with Epitech (Application number: 20150057269), along with academic investigators from a different institution. The subject matter of said patent (which deals with compositions and methods for the modulation of specific amidases for *N*-acylethanolamines for use in the therapy of inflammatory diseases) bears no relation to the present manuscript. The authors declare no potential conflict of interest with respect to the authorship and/or publication of this article.

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